

27. Answer: C

$$\begin{aligned} I_{\text{r.m.s.}} &= \frac{1}{2} I_0 \\ &= \frac{1}{2} \times 7.6 \\ &= 3.8 \text{ A} \end{aligned}$$

$$\begin{aligned} \text{Mean power dissipated } \langle P \rangle &= (I_{\text{r.m.s.}})^2 \times R. \\ &= (3.8)^2 \times 9.4 \\ &= 136 \text{ W} \approx 140 \text{ W} \end{aligned}$$